

Spectral Wave Transformation over an Elongated Sand Shoal off South-Central Louisiana, U.S.A.

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ABSTRACT

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Ship Shoal is an elongated sand shoal located along the 10 m isobath in south-central Louisiana adjacent to a rapidly eroding barrier island complex, Isles Dernieres. High quality sand resources from this shoal are considered as a viable source for the long term maintenance of the adjacent barrier islands. Previous wave modelling studies suggest that the shoal acts as a submerged barrier and mitigates storm waves. The present investigation employs a fully spectral finite element model to estimate the wave transformation over the shoal during a cold front generated storm event in April, 2005. MIKE 21, a spectral wave model, was implemented for the eastern Ship Shoal area at a high resolution scale, to estimate wave attenuation over the shoal and to better understand the directional spectrum when the storm-generated waves cross the shoal. The fine resolution coastal model was nested within a regional wave model for the Gulf of Mexico. NCEP re-analysed wind data were used as input and the spatial distribution of bottom sediments were also included in the model to select an appropriate friction factor. It was found that southerly storm waves could lose as much as 22% of their offshore wave height while propagating over the shoal. This level of wave energy reduction also points to the effectiveness of the shoal in shielding the already vulnerable coast against the frequent cold fronts and hurricanes. The model results were validated with time series data collected from *in situ* measurements on the shoal as well as using data from an observation site CSI 6. The dissipating wave energy over the shoal helps in resuspension and transport of shoal material during storm events.

ADDITIONAL INDEX WORDS: *cold fronts, wave climate, wave attenuation, significant wave height, numerical modelling, Ship Shoal, Louisiana*

INTRODUCTION

The Louisiana coast has undergone severe erosion over several decades (WILLIAMS ET AL., 1997) and a plausible solution put forward to mitigate this problem is to replenish the eroding barrier islands using high quality sand pumped from offshore shallow shoals. One such source, Ship Shoal, is located on the 10 m isobath in south-central Louisiana adjacent to a rapidly eroding barrier island complex, Isles Dernieres (Figure 1). The wave transformation characteristics that can be attributed to this elongated shallow shoal require further quantification. STONE ET AL., (2004) reported that the shoal transforms the approaching waves during storm conditions and that the removal of the shoal would significantly increase wave energy along the leeward flank of the deposit. Not much has been studied on the spectral evolution of the waves in the region due to the presence of the shoal, especially during high energy events associated with winter and spring cold fronts.

As the waves shoal in shallow waters, wave energy spectra evolve due to refraction, nonlinear energy transfers to higher and lower frequencies (ELGAR ET AL., 1990) and energy dissipation caused by wave breaking and bottom friction (THORNTON AND GUZA, 1983; SHERMET AND STONE 2003). The rapid shift in the wind and wave direction, especially when a cold front passes over

the region, significantly influences the wave spectral evolution. The present study is focussed on modelling the spectral wave transformation over the eastern half of Ship Shoal during a spring cold front event. MAA ET AL., (2004) also used similar modelling methods to study the impact of sand mining on the wave climate offshore of the Maryland and Delaware coasts.

NUMERICAL MODEL

In order to resolve the characteristic scales of the physical processes in the coastal waters, a fine mesh is required. A high resolution computational grid is also needed to resolve the complex bottom topographies in shallow water environments, such as barrier islands, shallow shoals and submerged bars (SORENSEN ET AL., 2004). The need for high resolution local models can be achieved using nesting techniques, where a local model with a fine mesh is embedded in a coarse mesh model.

MIKE 21 Spectral Wave (SW) model is used for the present study, which has also got nesting capabilities. This model was developed by the Danish Hydraulic Institute and was implemented successfully for modelling coastal wave characteristics of the North Sea (SORENSEN ET AL., 2004). Also, the model was applied successfully in San Francisco Bay, USA, to study changes in wave climate associated with a runway expansion project (KERPER, D.,

and Pers. Com.). The model is based on unstructured meshes and it simulates the growth, decay and transformation of wind generated waves and swells in offshore and coastal areas. The discretisation in geographical and spectral space is performed using a cell-centred finite volume method. In the geographical domain, an unstructured mesh is used. The integration in time is based on a fractional step approach (SORENSEN ET AL., 2004).

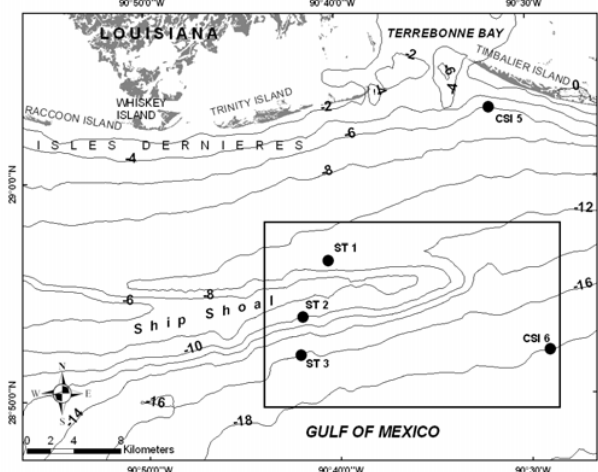


Figure 1 Study area. ● showing *in situ* observation stations at Ship Shoal and inshore

The model domain is the eastern section of Ship Shoal and extends over an area of $26.25 \times 13 \text{ km}^2$, the rectangular box inside Figure 1. The generated flexible mesh grid (Figure 2) had a spatial resolution of $\sim 480 \text{ m}$. A finer mesh was embedded in this coastal domain to further resolve the wave conditions across a transect of the shoal. The grid resolution for the finer mesh approximated 220 m . The coastal model was nested with an operational regional wave model for the Gulf of Mexico (JOSE AND STONE, 2006). For this larger domain, along the northern Gulf coast, the grid resolution was $\sim 2 \text{ km}$ while for the rest of the boundary a

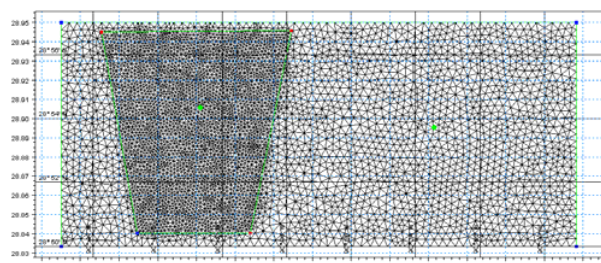


Fig. 2 Flexible mesh grid for the model domain

coarser grid of $\sim 30 \text{ km}$ was used. Fine scale bathymetry data (6 arc-second resolution) were used for the northern Gulf and coarse bathymetry for the remainder of the basin. The bathymetry data were referenced to Mean Lower Low Water (MLLW). The data used were extracted from the database provided by the National Geophysical Data Centre (NGDC), National Oceanic and Atmospheric Administration (NOAA).

Cold Fronts Along The Northern Gulf Coast

HSU (1988) and ROBERTS ET AL., (1989) provided detailed explanations for the mechanism of cold front generation,

sustenance and dissipation in the northern Gulf of Mexico. Once a cold front passes over the northern Gulf coast, wind direction changes from the southern quadrant to the north, wind speed increases and air temperature drops (KOBASHI ET AL., 2005). The northern Gulf of Mexico experiences frequent cold front events between October and May and they have played a critical role in the coastal hydrodynamics of the Gulf coast of the United States (STONE, 2000).

The wave simulations were carried out for a cold front generated storm event during 5th-10th April, 2005. During this period the maximum sustained wind speed and significant wave height measured at a coastal station, CSI 6 (see Figure 1), maintained by Louisiana State University (STONE ET AL., 2003), was 14.2 m/s and 2.1 m respectively. Input for the model, reanalysed wind data, in a grid format, were extracted from the National Centre for Environmental Prediction (NCEP), NOAA, database. The North American Regional Re-analysed (NARR) model grid has a horizontal resolution of 32 km and covers the entire continental US and the Gulf of Mexico region. The u and v components of the wind vector for the Gulf of Mexico region were extracted from the regional grid using Matlab^R routines. For comparison, the time series data of measured and re-analysed wind speed and wind direction at the CSI 6 station are presented in Figure 3.

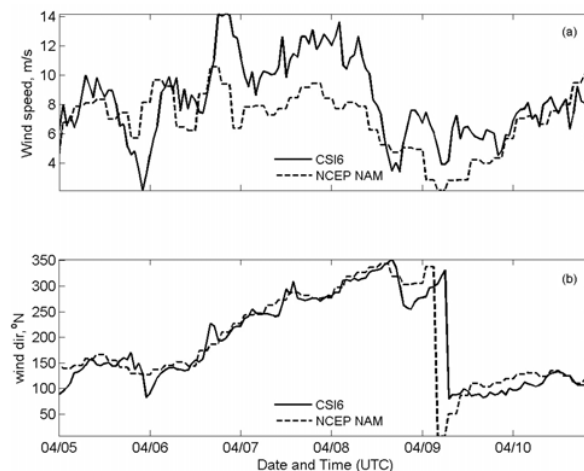


Figure 3 (a) wind speed and (b) wind direction at the CSI 6 station. NCEP NARR model wind data (broken curve) plotted against the measured data.

Tidal elevation data for the Wine Island, Timbalier Bay, along the Louisiana coast, were extracted from the NOAA tide prediction tables. The tidal data were also referenced with respect to MLLW. The maximum tidal range observed during the study period was 0.48 m , on 5th April, 2005. The interpolated data (Figure 4) were included in the coastal model to define the water surface elevation. Median grain size (D_{50}) distribution data for the Ship Shoal region were collected from the United States Geological Survey (USGS) database, usSEABED, and were used for the bottom friction parameter.

FIELD DATA

This study was conducted in association with a field survey program along eastern Ship Shoal during spring 2005. Three stations were set up across a transect of the shoal (Figure 1) and an array measuring wave, current and turbidity was deployed for a period of 34 days in April-May, 2005. Data from an Acoustic

Doppler Velocimeter (ADV), deployed at the northern Station ST1, was used for model skill assessment. Also, time series data collected from the CSI 6 station (Figure 1) were used for the model validation.

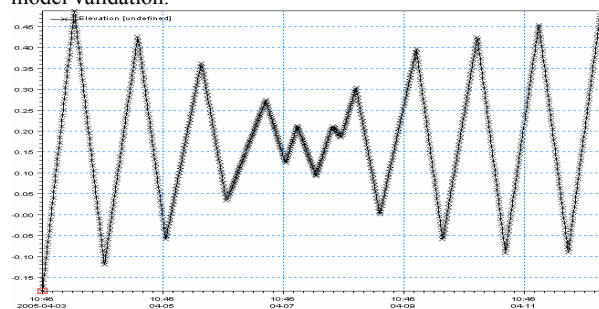


Figure 4. Tidal elevation data for Wine Island and Timbalier Bay during the study period.

SIMULATIONS AND SKILL ASSESSMENT

A fully spectral instantaneous approach was used for the computation of the wave parameters. A logarithmic frequency discretisation with 25 frequencies was used. The lowest discrete frequency was $f_{\min} = 0.04$ Hz and the ratio between successive frequencies was chosen as 1.1. The number of discrete directions was chosen as 16. The time step interval chosen for the simulation was 75 s. The white capping parameters were included in the model ($C_{\text{dis}} = 2$ and $\Delta_{\text{dis}} = 0.8$). For the wave breaking parameter, a constant value of $\gamma = 0.8$ was used. The model computed the wave parameters using the forecast wind input. Synoptic maps of significant wave height (H_s) (Figures 5 & 6), wave period, wave direction etc. for the study site were generated. For the purpose of skill assessment of the model, time series outputs were also generated for the 3 stations where the Oceanographic survey was conducted. The stations ST1, ST2 and ST3 were at depths of 11.6 m, 12.8 m and 16.2 m respectively (Figure 1). Outputs were also generated for the CSI 6 station, located at the eastern end of the domain.

The wave fields associated with the storms generated by the approaching cold front is given in Figure 5. At the beginning of the simulation the waves were from the south in line with the prevailing wind direction and the maximum wave height computed was 1.5 m. The higher waves were concentrated along the southwestern corner of the domain where the depth-induced shoaling enhanced the wave heights. As the cold front crossed the study area the wind direction rotated and hence the wave fields also switched to the north and north-west (Figure 6). The maximum wave height observed during this time was 0.6 m. The wave energy attenuation along the crest of the shoal during this wave rotation period can also be deciphered from Figure 6.

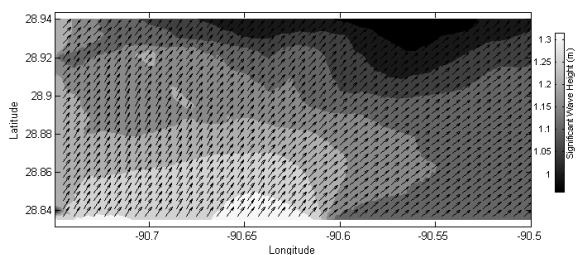


Figure 5. Simulated wave fields corresponding to the approaching phase of the cold front.

The simulated significant wave height, peak wave period and mean wave directions for the three stations are plotted along with measured wave parameters from northern station, ST1 (Figure 7). Measured data were smoothed using a 5 hour moving average filter. The simulated significant wave height are in good agreement with the measured data except during the peak storm period, prior to the reversal of the wind and wave direction (Figure 7a). Time series of peak wave period also shows good correlation

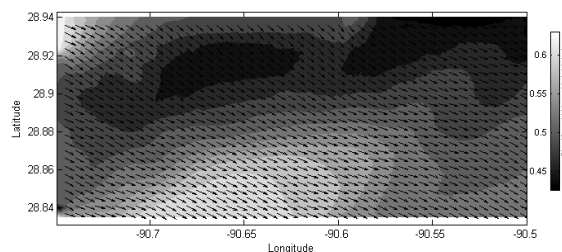


Figure 6. Simulated wave fields when the cold front crossed the study area

with the measured data (Figure 7 b). Mean wave direction shows good agreement with the measured data except during the reversal of the wave directions (Figure 7 c), when the simulated wave direction remained northerly for a prolonged period of time.

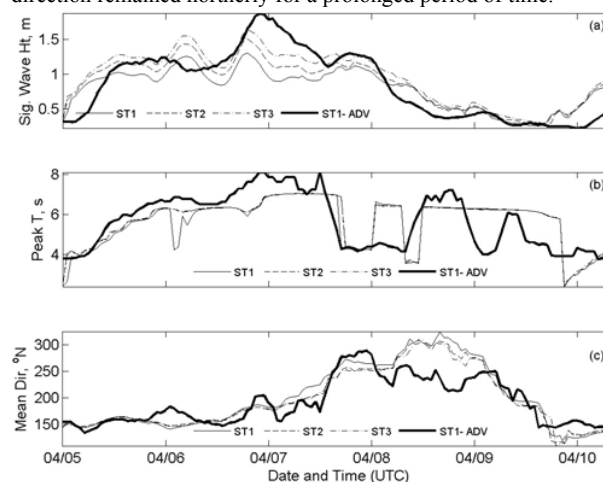


Figure 7. Time series of (a) significant wave height, (b) peak wave period and (c) mean wave direction, simulated for the three reference stations. The thick line corresponds to the measured data from station ST1.

Model outputs are also compared against the measured data from station CSI 6 (Figure 8). Simulated significant wave height values are in good agreement with the measured data (Figure 8a) except during the peak storm activity when the model under predicts the wave height. The plot of Peak wave period distribution (Figure 8b) shows a very strong correlation between measured and model output.

The discrepancy in the model outputs with that of the measured data can be attributed to the resolution and accuracy of the input wind data. It is clear that wave models are extremely sensitive to wind inputs. For a fully developed sea, sensitivity experiments revealed that small errors in the input wind can result in considerable differences in the computation of wave parameters (RAJ KUMAR ET AL., 2000). It can be observed in Figure 3a that the NARR model wind data generally under predicts the measured data from station CSI 6. However, for the offshore locations, the

model winds are generally in good agreement with the measured data. JOSE AND STONE (2006) reported very strong correlation between the NARR model wind data and measured data from the National Data Buoy Centre (NDBC) buoys off the Louisiana coast.

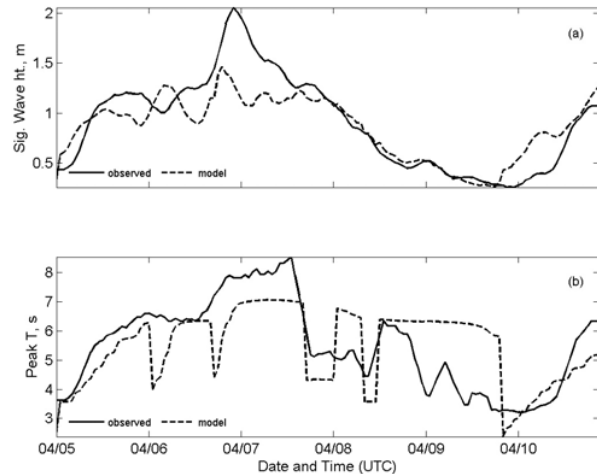


Figure 8. Time series of (a) significant wave height and (b) peak wave period at CSI6 station.

Wave height and spectral transformation over the shoal

The Percentage of attenuation of significant wave height and peak wave period associated with waves crossing the shoal were computed between the three stations (Figure 9). As expected, the maximum wave height attenuation occurred between the offshore station and the nearshore station (ST1-ST3). The highest percentage of attenuation calculated was 22.12 for the southerly waves and 27.68 for the northerly waves (Table 1). Between stations ST2 and ST3 the corresponding values are 15.67 and 24.13. Between ST1 and ST2 the percentage attenuation values are 15.85 and 13.87 respectively. It is also observed that between stations ST2 and ST3 there is no significant transformation for the peak wave period (Figure 9b) while for the ST1-ST2 and the ST1-ST3 stations, the peak wave period also undergoes transformation throughout the study period.

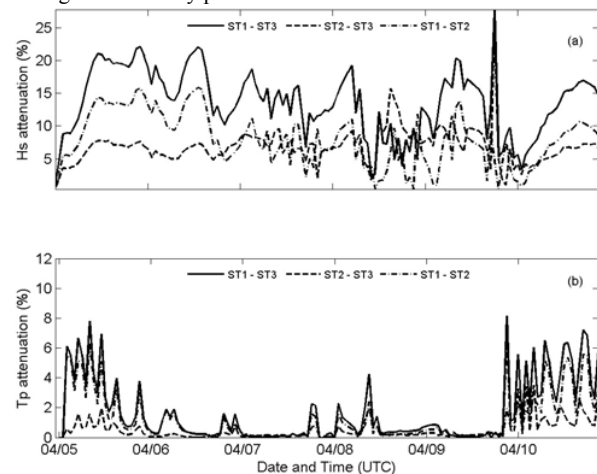


Figure 9. Percentage of attenuation of (a) significant wave height and (b) peak wave period, across the three stations.

TABLE 1. Percentage of attenuation of significant wave heights across the shoal.

Station Pair	Distance (km)	Percentage of attenuation	
		southerly waves	northerly waves
ST3-ST1	8.37	22.12	27.68
ST3-ST2	3.22	15.67	24.13
ST2-ST1	5.29	15.85	13.87

The evolution of wave energy spectra during a cold front generated storm is given in Figure 10. Before the cold front passes the study area the waves are from the south south-easterly direction and the directional spectrum shows an almost similar energy distribution pattern for all three stations (Figures 10, a, c & e). Once the cold front passes the region, a complete rotation of the wave spectra occurs, migrating from south-southeast to north-northwest. It is observed that the energy spreads across a wide range of frequency and direction, following the rapidly shifting wind conditions during this transition phase. For the inshore station (Figure 10b) the peak wave energy is centred at the northerly direction and the spectra spread from 45° to 225° . For the middle station, ST2, the wave spectrum is distinctly bimodal

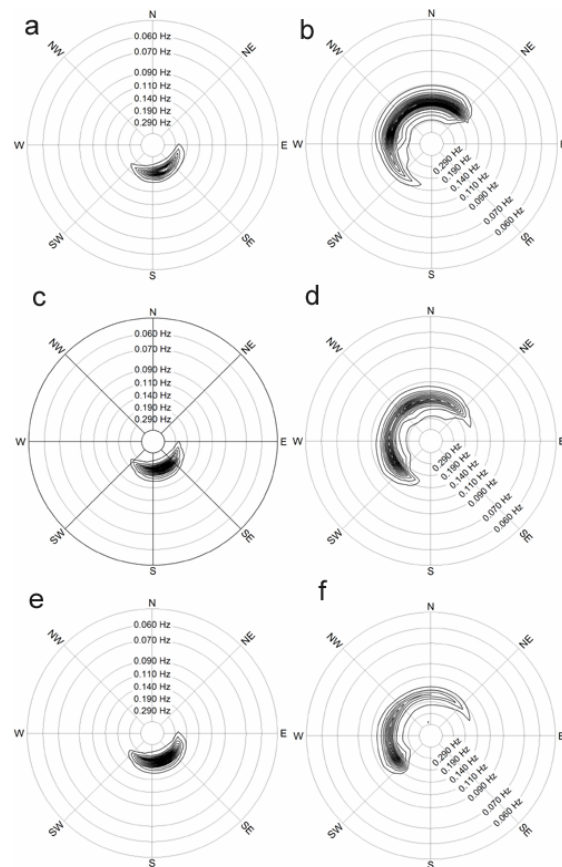


Figure 10. Evolution of spectral wave energy at three stations (a & b) station ST1, (c & d) station ST2 and (e&f) station ST3.

(Figure 10 d) with dominant directions centred at 247.5° and 320°. For the offshore station, ST3, the dominant direction is rotated further to the south (Figure 10 f) and centred at 225°. This shift in dominant direction of the energy spectra during the passage of the cold front is attributed to the veering of wind direction from southwest to north as well as to the refraction of the waves over the elongated shoal. More field observations across the shoal are required to further explain this wave transformation process.

CONCLUSION

The wave fields generated by a cold front induced storm were simulated using a third-generation spectral wave model. Using predicted wind data as input spectral and time series wave parameters were computed and skill assessed using measured data from two locations. Agreement between the model results and the observed data is found to be good except during the peak storm conditions. This discrepancy is attributed to the resolution and the accuracy of the input wind data.

The percentage of attenuation of significant wave height and peak wave period, as the waves cross the shoal, was computed. Between the two flanks of the shoal a significant wave height attenuation of 22.12% was computed for the southerly waves and 27.68% for the northerly waves. The dissipation of wave energy over the shoal results in resuspension and transport of shoal material during storm events. KOBASHI ET AL. (2006) observed significant bed level changes along eastern Ship Shoal in association with a spring cold front event. This level of wave energy attenuation also points to the effectiveness of the shoal in shielding the already vulnerable coast against frequent cold fronts and occasional hurricanes. The evolution of the energy spectra, when the cold front crossing the study area is quite abrupt and the dominant direction shifts from southwest to north across the shoal. This shift is attributed to wave refraction due to the shoal as well as the veering of the wind direction.

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